YANDING

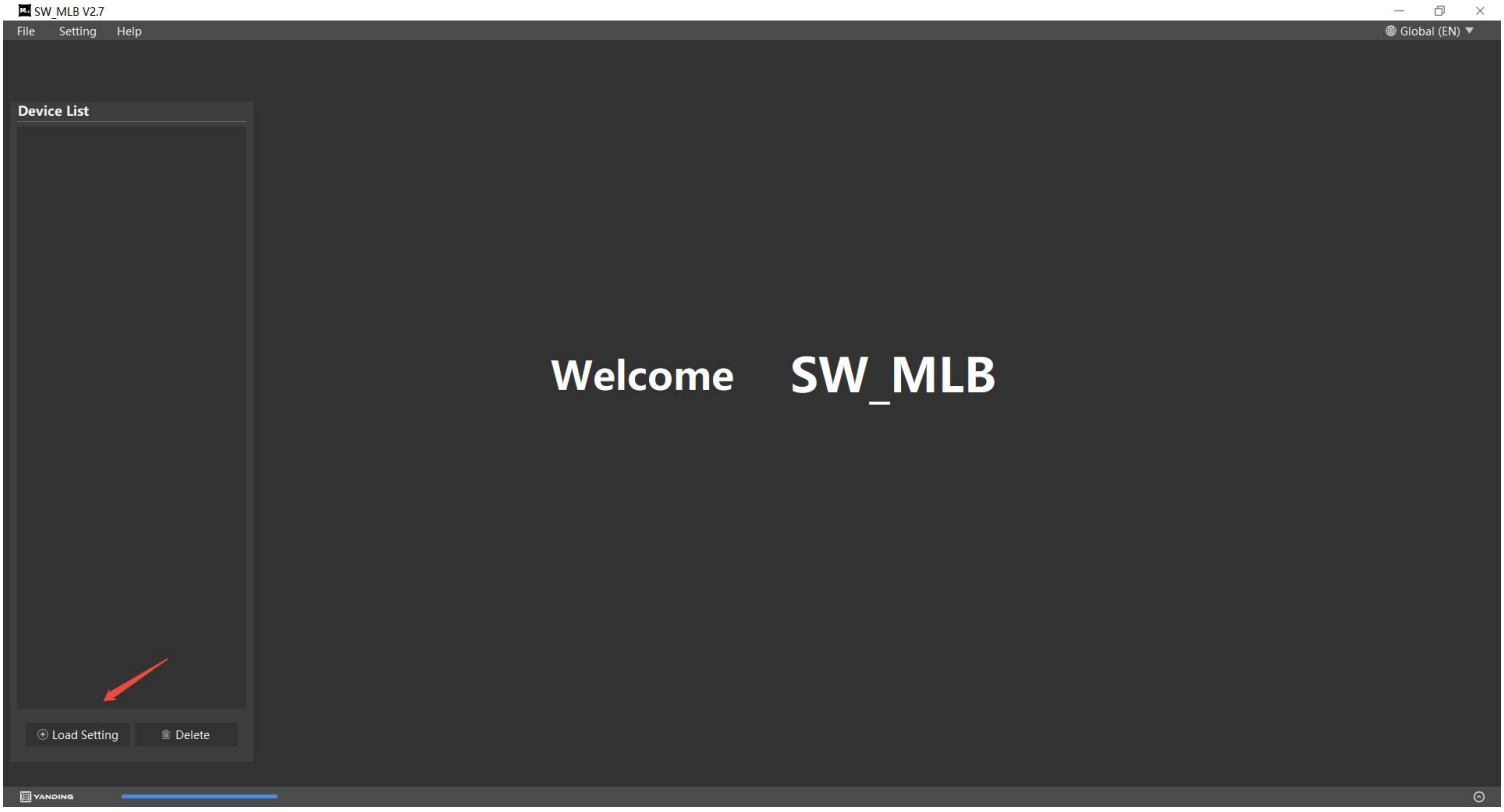
Real-scene Light Box Testing System User Manual

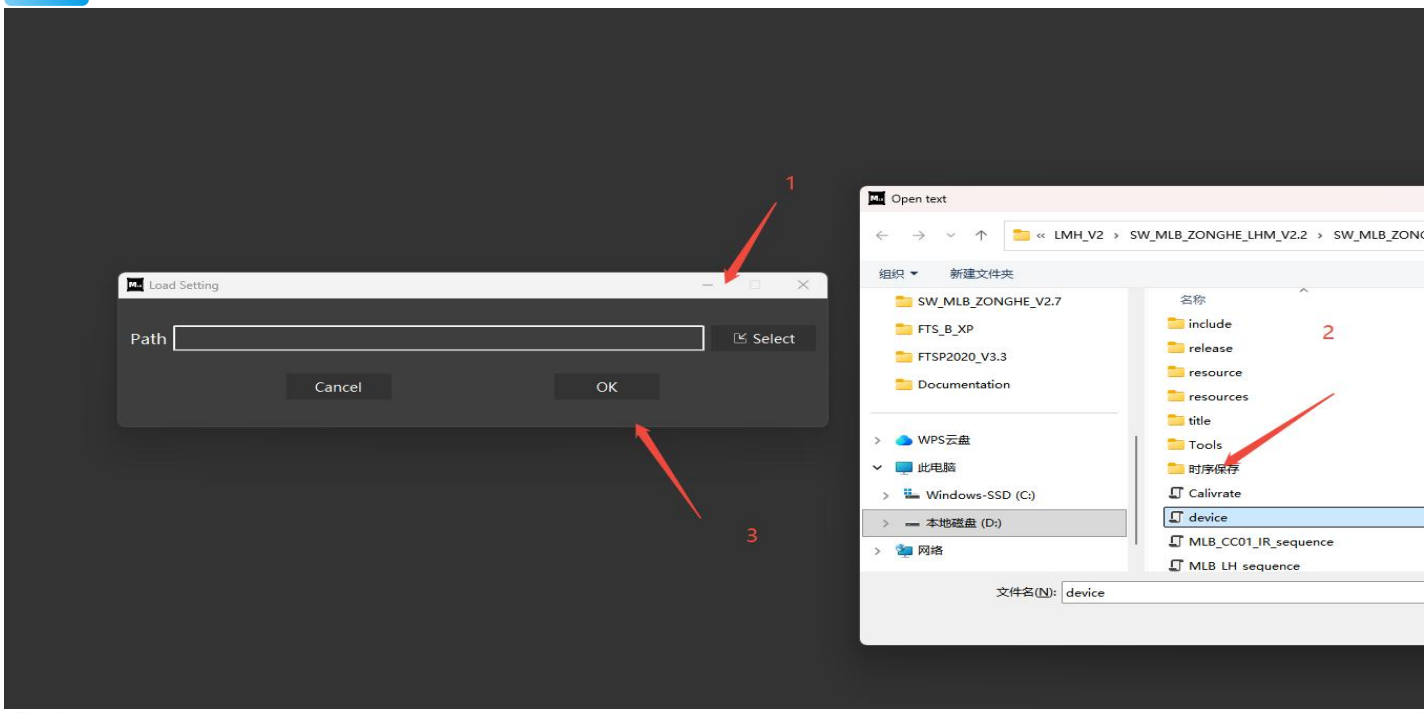
Email: sales@rdbuy.com
Website: www.yanding.com

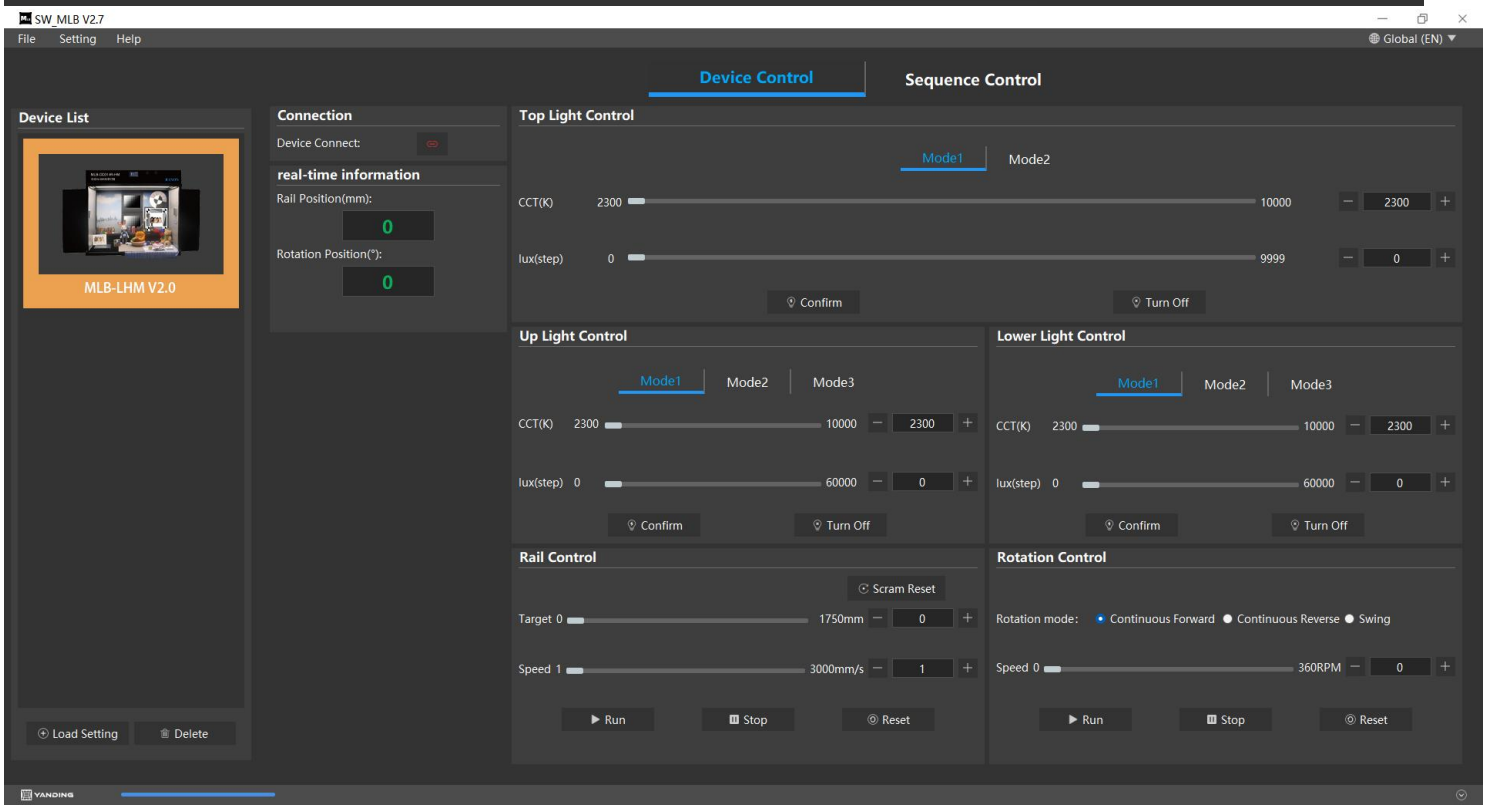
Software Configuration Introduction

1. Device Import Operation

As shown in the figure, click 'Load Configuration File' for the device, select the corresponding device.json file to import the device, check the desired device on the left side, and then click 'Confirm' to load the corresponding device control page.







邮箱： sales@rdbuy.com 网址： www.yanding.com

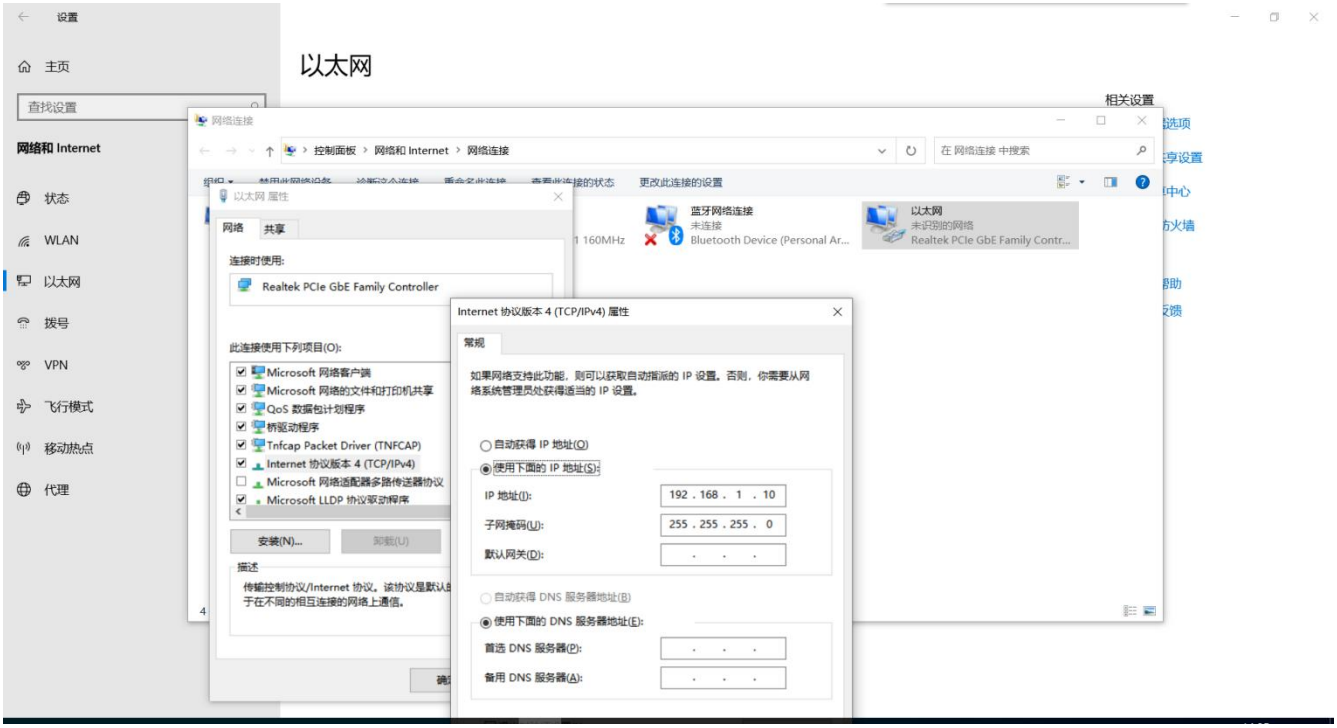
Connect

2.1 Preparation Work

Connect the controller to the computer using a USB cable. Turn on the device's power switch and wait for initialization to complete. Then configure the Ethernet IP address according to the "IP Configuration Process Guide."

2.2. IP Parameter Configuration

As shown in the figure, after opening the computer settings, select Network and Internet Options, click More Network Adapter Options, locate the Ethernet adapter created for the connected device, double-click it, find the TCP/IPv4 options, change the IPv4 address to 192.168.1.* (use a larger number whenever possible), and confirm.

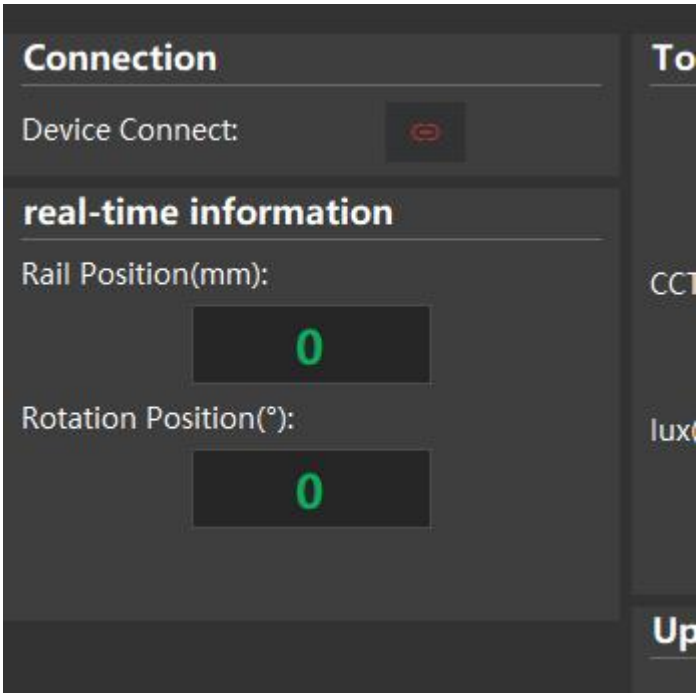


2.3 Software connection and log display

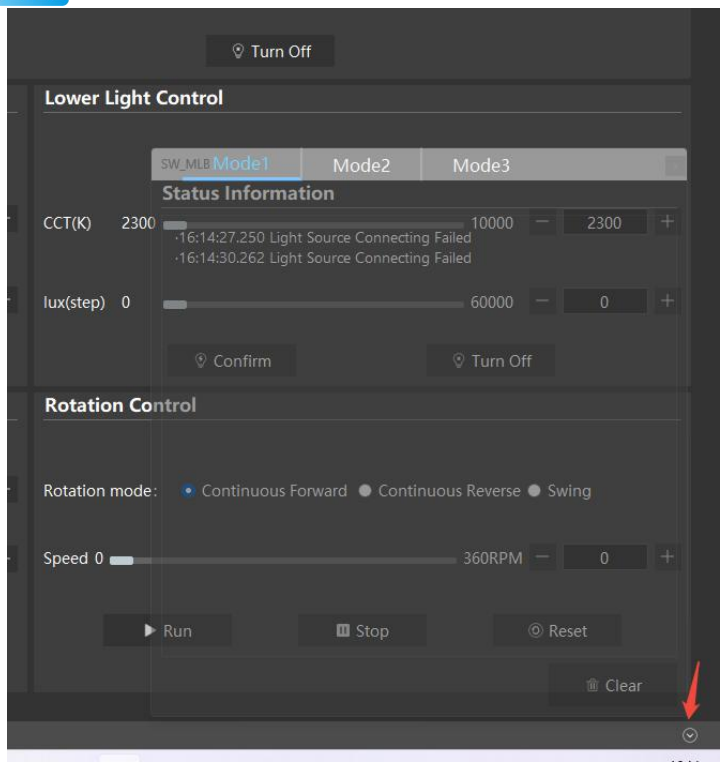
Click the device icon in the device list to select and control the device. When the indicator light in the left connection area turns green, the device is connected; red indicates no connection. The real-time information area displays the current

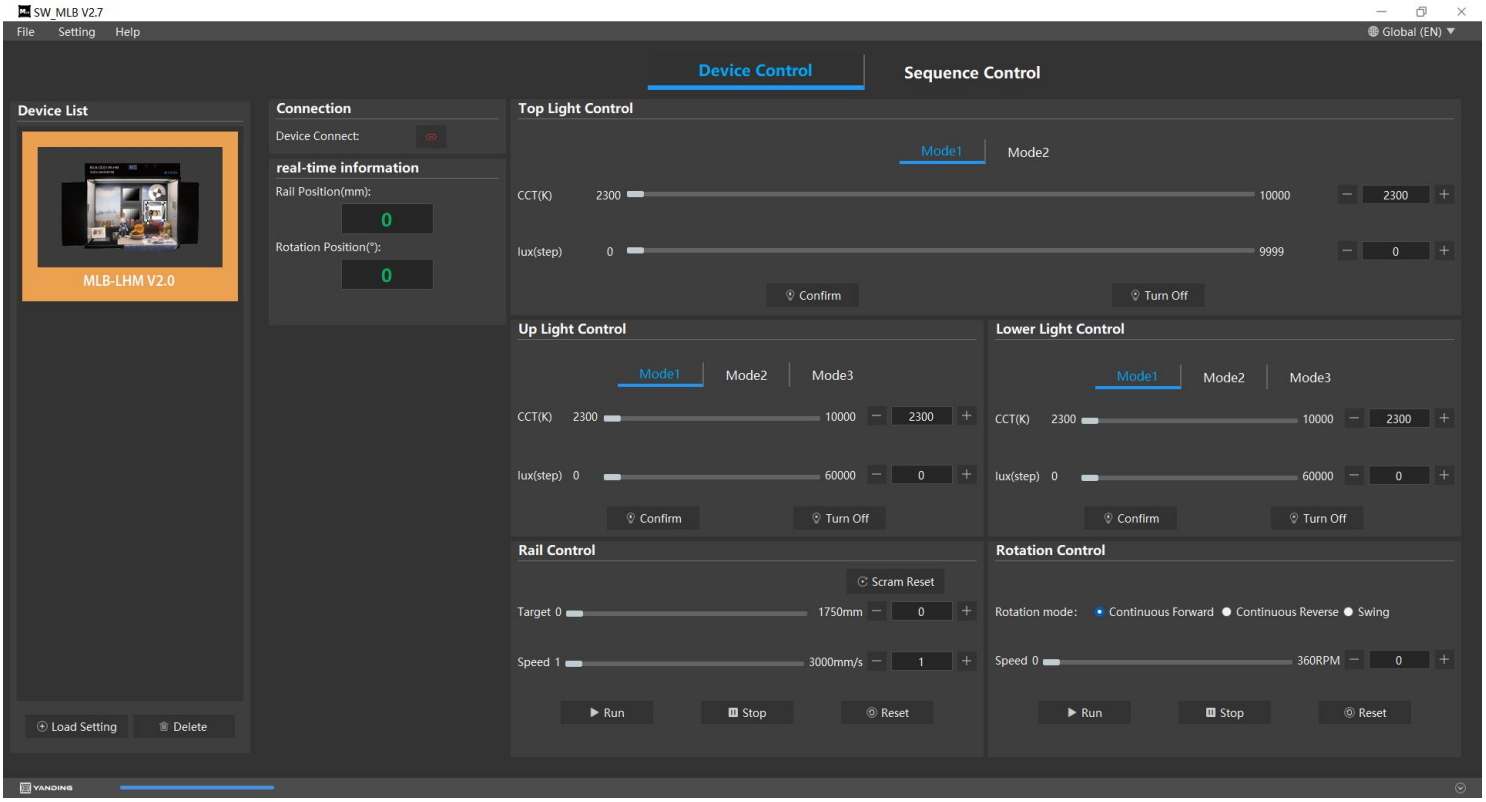
邮箱： sales@rdbuy.com 网址： www.yanding.com

rail position (mm) and the current turntable position (°).



Click the icon in the lower-right corner of the interface to open the log window for viewing operation logs; it is initially closed. Note: If the connection fails, check that the line and configuration are correct, or click the Load Configuration File button in the lower-left corner to re-scan the device.





邮箱： sales@rdbuy.com 网址： www.yanding.com

3. Device Control

The device control page is used for manual debugging, single-parameter configuration, and functional verification of the equipment, serving as an essential preparatory step before time-series programming.

3.1 Top Light Source Control

The top light source is a surface light source supporting two operating modes: Mode 1: White light mode – adjust color temperature and illuminance; Mode 2: Infrared light mode – adjust the intensity of 850 nm and 940 nm infrared light. Operation steps: Click the Mode 1 or Mode 2 button to switch modes. Mode 1 parameters: Color temperature (K): Drag the slider or enter a value directly (range: 2300K – 10000K); Illuminance (steps): Drag the slider or enter a value directly (range: 0 – 60000 steps). Click the OK button to apply the parameters and activate the top light source; click the Off button to extinguish it.

3.2 Upper/Lower Light Control

The upper and lower lights are symmetrical strip-shaped light sources, supporting three operating modes: Mode 1: White light mode – adjust color temperature and illuminance; Mode 2: Infrared light mode – adjust the intensity of 850 nm and 940 nm infrared light; Mode 3: Flicker mode – adjust flicker frequency, duty cycle, and flicker illuminance.

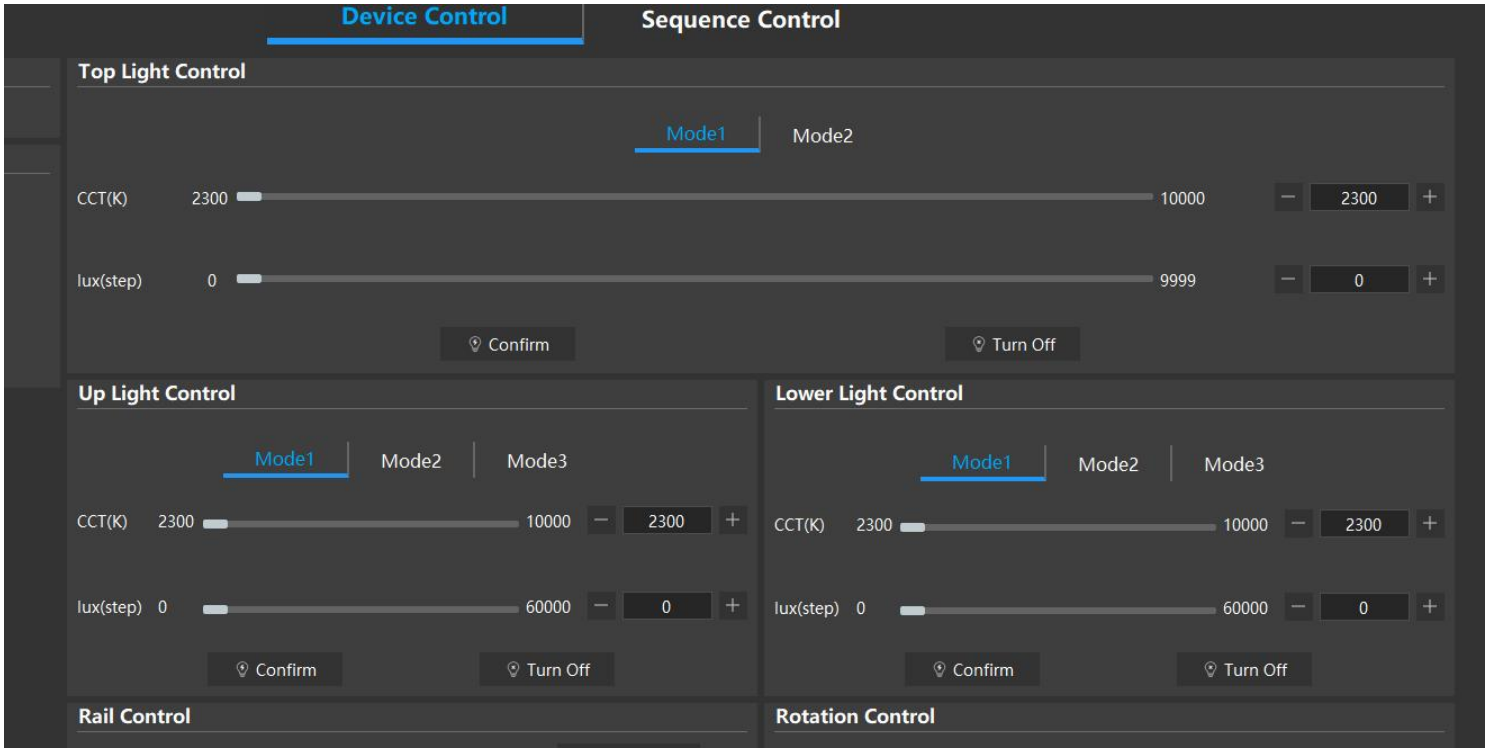
Operation steps: Click the Mode 1/Mode 2/Mode 3 buttons to switch operating modes. Set the corresponding parameters according to the selected mode:

Mode 1: Color temperature (2300K – 10000K), illuminance (0 – 60000 steps) Mode 2: 850 nm intensity (0 – 60000 steps), 940 nm intensity (0 – 60000 steps) Mode 3: Frequency (0 – 1000 Hz), duty cycle (1 – 100%), flicker illuminance (0 – 60000 steps)

Click the Confirm button to apply the parameters and activate the corresponding light source

Click the Close button to turn off the corresponding light source

✓ Note: The upper and lower lights are fully independent and can be set to different modes and parameters.



3.3 Guide Rail Control

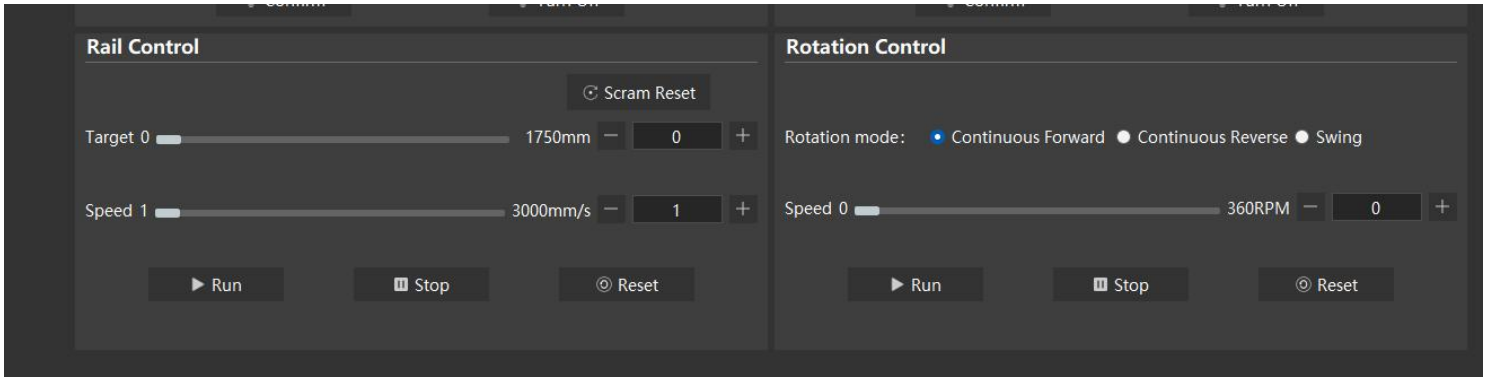
The guide rail moves rapidly forward and backward to simulate various testing scenarios. Operating steps: Parameter settings: Movement distance (mm): Enter the target position, range 0 - 1750 mm; Movement speed (mm/s): Drag the slider or enter a value, range 1 - 3000 mm/s. Click the Run button to move the guide rail to the specified position.

Click the Stop button to immediately stop the rail movement

3.4 Wheel Control

The turntable is designed for rotation to simulate various testing scenarios involving directional rotation and oscillation. Operation steps: Rotation mode selection: Continuous forward rotation: The turntable rotates continuously clockwise. Continuous reverse rotation: The turntable rotates continuously counterclockwise. Oscillation: The turntable swings back and forth within a specified angle range. Rotation speed setting: Drag the slider or enter a value

(range: 0 - 360 RPM). Oscillation speed range: 0 - 10 levels. Click the Run button to start the turntable. Click the Stop button to immediately halt the turntable's movement.



4. Time Sequence Control (Automated Testing)

The timing control page is the core functionality of the system, enabling visual editing of synchronized timing across multiple devices and facilitating fully automated testing workflows.

4.1 Basic Concepts of Time Series

A sequence consists of multiple steps, each defining the state of a device within a specific time period. The system executes all steps in chronological order and supports cycling through the entire sequence.

4.2 Light source timing settings

The system supports three independent light source sequences: top light source sequence, upper light sequence, and lower light sequence. Each sequence can include multiple steps. Steps: Click the "Add" button on the right side of the corresponding light source sequence table to add a new step. Double-click a cell in the table to edit the step parameters: Table

Parameter Name	explain	Applicable Mode

Parameter Name	explain	Applicable Mode
Light Source Mode	Select the light source mode for this step (1/2/3)	All Modes
colour temperature K	Color temperature value in white light mode	pattern 1
illuminance lv	Number of illumination levels in white light mode	pattern 1
850nm st	850 nm infrared light intensity	pattern 2
940nm st	940 nm infrared light intensity	pattern 2
Frequency Hz	flicker frequency	pattern 3
Duty %	Flash duty cycle	pattern 3
Flash illuminance lv	Illumination value in flash mode	pattern 3
start time(S)	The time point when this step begins execution (unit: seconds)	All Modes
terminal time (S)	The time point when this step ends execution (unit: seconds)	All Modes

Click the **×** button to the right of the step to delete it

Repeat the above steps and add all required light sources.

Tip: The time ranges of the steps can overlap to achieve simultaneous operation of multiple light sources.

4.2 Exercise Sequence Settings

The motion sequence controls the synchronized movement of the guide rail and turntable, allowing parameters for both to be set simultaneously for each step.

邮箱： sales@rdbuy.com 网址： www.yanding.com

Steps: 1. Click the **Add** button on the right side of the motion sequence table to add a new step. 2. Double-click a cell in the table to edit the step's parameters.

table

Parameter Name	explain
distance mm	Guideway target position (0 - 1750 mm)
velocity mm/s	Guideway movement speed (1-3000mm/s)
Rotation Method 1	Rotating mode of the turntable (continuous forward rotation / continuous reverse rotation / oscillation)
velocity	Turntable rotation speed (0 - 360 RPM) Swinging speed (0 - 10 g)
Single rotation time	Single rotation time in the oscillating mode

Click the **×** button to the right of the step to delete it. Repeat the above steps to add all required exercise steps.

4.3 Global Time Setting

Loop Count: Set the number of loop executions for the entire sequence. Enter 1 to execute once. Clear All: Click this button to erase all steps in the sequence table. Export Settings: Click this button to save the current sequence configuration as a JSON file (.mlb format). Import Settings: Click this button to load the saved sequence configuration file.

4.4 Execution Sequence

Check that all timing parameters are correctly set and the device is properly connected. Click the Run button in the lower-right corner to initiate timing execution. The number of executions will display the current completed loop count in real time. To pause the timing mid-execution, click the Stop button.

⚠ Note: Do not manually operate any buttons on the device control panel during timing execution to avoid disrupting the timing sequence.

